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Volatility and co-movement between ESG and non-ESG futures and uncertainty indices: Evidence from wavelet coherence and GARCH models

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ABSTRACT

This study examines the dynamic relationship between environmental, social, and governance (ESG) futures and conventional stock index futures, as well as their co-movement with key uncertainty indices i.e. the CBOE Volatility Index (VIX), Geopolitical Risk Index (GPR), and Economic Policy Uncertainty Index (EPU) over the period 2021–2025. Using wavelet coherence and GARCH models, the analysis captures time-frequency dependence and volatility persistence under varying market conditions, including the aftermath of the COVID-19 pandemic, the war in Ukraine, and the 2025 U.S. tariff shock. The results reveal strong positive correlations between ESG and non-ESG futures, indicating a high degree of integration across markets. Among the uncertainty measures, the VIX exerts the most consistent and significant influence, with pronounced co-movement across multiple time scales. While ESG futures exhibit volatility clustering similar to their conventional counterparts, differences in persistence point to a form of relative resilience during turbulent periods. However, this resilience does not reflect safe haven properties. Instead, ESG futures remain strongly exposed to systemic risk factors and respond to shocks in a manner comparable to traditional assets, particularly under conditions of heightened financial market uncertainty. The findings further demonstrate that the relationship between futures and uncertainty is heterogeneous and depends on the nature of the underlying shock. Financial market volatility induces immediate and synchronized responses, whereas geopolitical and policy-related uncertainty operate through weaker or more gradual transmission channels. This study contributes to the literature on sustainable finance by providing new evidence on the behaviour of ESG derivatives in a time–frequency framework and by clarifying their role within the broader system of financial risk transmission. The results offer practical implications for investors and policymakers, suggesting that ESG futures may provide differentiated volatility dynamics but should not be considered as protective or hedging instruments in periods of systemic stress.

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1. Introduction

In recent years, the financial markets have witnessed a growing interest in sustainable investing, with environmental, social, and governance (ESG) indices gaining importance alongside conventional benchmarks. ESG futures, in particular, have emerged as instruments that allow investors to hedge, speculate, and diversify portfolios while aligning with sustainability principles. Despite their increasing popularity, it remains unclear whether ESG derivatives exhibit different risk and return dynamics compared to their traditional counterparts, especially in periods of heightened uncertainty.

Uncertainty plays a pivotal role in shaping financial market behaviour. Measures such as the CBOE Volatility Index (VIX), the Geopolitical Risk Index (GPR), and the Economic Policy Uncertainty Index (EPU) are widely employed to capture different dimensions of financial, political, and policy instability. These indices have been shown to capture the extent to which market uncertainty and instability influence asset pricing, thereby enabling an examination of how prices reflect such uncertainty. However, limited attention has been paid to their interaction with ESG derivatives markets, particularly in the context of extreme events such as the COVID−19 aftermath, the war in Ukraine, and the introduction of U.S. tariffs in 2025.

The existing literature provides mixed evidence on the resilience of ESG investments during periods of financial turmoil. Several studies suggest that ESG-oriented assets demonstrate greater stability and hedging potential in times of heightened uncertainty. For example, (Piserà and Chiappini, 2024) find that ESG indices in China served as effective hedging instruments during the COVID−19 crisis, although they did not exhibit strong safe-haven properties. Similarly, (Rubbaniy et al., 2021) show that the safe-haven features of ESG assets are conditional and depend on the prevailing level of market fear and uncertainty. In contrast, other research indicates that the risk-return profile of ESG portfolios remains closely aligned with conventional markets. While (Albuquerque et al., 2020) document that companies with higher environmental and social scores experienced higher returns and lower volatility during the pandemic, (Cagli et al., 2022) reveal that ESG indices can act as transmitters of volatility to commodity markets, suggesting limited diversification benefits. Furthermore, recent time–frequency analyses (Wan et al., 2024) highlight that the connections between global ESG and non-ESG indices vary across investment horizons, underscoring the complex and dynamic nature of their co-movements with uncertainty measures. This body of evidence calls for further systematic investigation of the dynamic linkages between ESG and non-ESG assets across multiple time scales.

To address this gap, this study applies wavelet coherence and GARCH modelling to a dataset of ESG and non-ESG stock index futures (STOXX Europe 600, EURO STOXX 50, S&P 500) covering the period 2021–2025. Wavelet methods enable the detection of co-movements in both time and frequency domains, while GARCH models capture volatility persistence and clustering. By combining these approaches, the paper contributes to the understanding of how ESG derivatives behave under stress and whether they offer distinct risk-return characteristics compared to conventional futures.

The main objectives of this research are as follows: (1) to examine the dynamic co-movement between ESG and non-ESG futures and uncertainty indices; (2) to evaluate whether ESG futures display differentiated volatility patterns; (3) to identify lead-lag relationships between futures returns and uncertainty measures; and (4) to assess the impact of major geopolitical and policy events on futures markets.

Based on the above objectives, the following research questions were formulated:

1. How do ESG and conventional futures co-move with uncertainty indices (VIX, GPR, EPU) across short-, medium-, and long-term horizons?
2. Does the volatility of ESG futures differ significantly from that of their traditional counterparts?
3. Do uncertainty indices lead or lag futures market dynamics?
4. How do major geopolitical and policy shocks shape volatility in ESG and non-ESG futures markets?

To address these research questions, the remainder of the paper is structured as follows. Section 2 presents a comprehensive review of the existing literature on ESG investments, derivatives, and uncertainty measures. Section 3 describes the dataset, variables, and data sources employed in the empirical analysis. Section 4 outlines the methodological framework, including the wavelet coherence approach and GARCH modelling strategy. Section 5 discusses the main empirical results, highlighting the dynamic relationships between ESG and non-ESG futures and the role of uncertainty indices. Finally, Section 6 concludes the paper by summarising key findings, outlining practical implications, and proposing directions for future research.

2. Literature Review

The debate on whether environmental, social, and governance (ESG) investments offer resilience during crises has attracted considerable scholarly attention. Several studies argue that ESG-labelled assets exhibit relative strength in periods of turbulence, either through lower drawdowns, more stable flows, or improved risk-adjusted returns (Belloni et al., 2020; Nagy and Giese, 2020). For example, some funds with high sustainability scores recorded net inflows during the early stages of the COVID−19 crisis, suggesting that investors perceived ESG as a safer asset class (de Renzis and Mosson, 2024; Fang and Parida, 2022; Pastor and Blair Vorsatz, 2020; Supatgiat et al., 2025). However, the evidence is far from conclusive. Other empirical analyses highlight that ESG portfolios suffered losses similar to conventional benchmarks during COVID−19 (Sharma et al., 2024; Zhang et al., 2022) or that outperformance was driven primarily by sectoral composition (e.g., overweighting technology, underweighting energy) rather than intrinsic ESG characteristics (Bruno et al., 2021; Mahmood, 2021). This mixed evidence suggests that ESG instruments cannot be assumed to be inherently more resilient. Instead, context, methodology, and time horizon matter.

Most empirical work on ESG concentrates on equity indices, mutual funds, and ETFs rather than derivative instruments. Meta-analyses and review papers document that the bulk of studies examine companies' outcomes or fund performance, leaving derivatives underexplored (Whelan et al., 2015). At the same time, exchanges have begun to productize ESG exposures: major venues list ESG index futures (for example, CME's E-mini S&P 500 ESG futures and Eurex/STOXX's STOXX® Europe 600 ESG-X and EURO STOXX 50 ESG futures), and trading activity and product specifications on exchange websites confirm these contracts are tradable and increasingly liquid, justifying closer academic attention to ESG derivatives (CME Group, 2025; EUREX, 2025; STOXX, 2021).

Studies that examine volatility in ESG indices typically rely on GARCH models to capture heteroskedasticity, volatility clustering, and persistence. For example, (Sharma et al., 2024) analysed Indian companies sorted by ESG score and showed that both high and low-ESG portfolios experienced volatility clustering during the COVID−19 pandemic with limited evidence of an ESG "volatility shield". (Górka and Kuziak, 2022) apply GARCH models to ESG and conventional index returns over 2007–2019 and find that volatility persistence is high for both types of indices, confirming that ESG indices exhibit the same heteroskedastic clustering behaviour captured by GARCH methods. (Gameel, 2025) uses a TGARCH model to analyse ESG stock indices in Egypt and Turkey (2020–2025) and finds that both ESG and conventional indices exhibit asymmetric volatility, i.e. negative shocks have a different impact than positive ones and varying persistence across short- and long-term horizons. Further extensions have combined GARCH frameworks with macroeconomic or uncertainty indicators. (Mishra et al., 2023) integrated ESG index volatility with LSTM-based models and confirmed the robustness of classical GARCH specifications in forecasting ESG-related risks. Overall, these studies highlight that ESG indices display volatility characteristics broadly comparable to traditional markets, although differences in persistence and asymmetry may emerge under crisis conditions, providing motivation for further analysis using advanced volatility models.

A second body of work examines how uncertainty indices influence asset prices and volatility. The VIX index is a forward-looking measure of implied volatility on the S&P 500 and has been widely used to capture short-term financial stress (Whaley, 2000). The Economic Policy Uncertainty (EPU) index was introduced by (Baker et al., 2016) quantifies policy-related uncertainty from media frequency counts and is shown to affect investment, returns, and volatility in equity and commodity markets. More recently, (Caldara and Iacoviello, 2022) developed the Geopolitical Risk (GPR) index, which tracks the incidence of newspaper references to geopolitical tensions. GPR spikes are empirically associated with contractions in investment, output, and increased market risk premia. Applications of these indices to green or ESG assets remain limited but growing. For example, (Dai et al., 2022) showed that global EPU predicts crude oil futures volatility using a GARCH-MIDAS framework, highlighting the usefulness of policy uncertainty in derivative market modelling. (Wei et al., 2022) use wavelet decomposition and quantile methods for green bond markets (2014–2021) to see how EPU relates across time-frequency bands and under different quantiles. The combination of VIX, EPU, and GPR thus provides a multidimensional approach to uncertainty, each capturing different transmission mechanisms.

In our paper, we employ wavelet coherence analysis, as standard econometric techniques such as correlation or regression often mask the fact that relationships between assets evolve across both time and frequency domains. The continuous wavelet transform and wavelet coherence have therefore become popular tools in financial econometrics, following the formalisation by (Torrence and Compo, 1998). These methods enable researchers to disentangle short, medium, and long-term co-movements and to determine whether lead-lag relationships vary over time. Recent applications in finance underscore the utility of wavelet coherence. For example, (Yousaf et al., 2024) examine the dynamic connections between a News Sentiment Index (NSI) and ESG leader indices of developed countries using time-frequency wavelet techniques, finding that relationships vary across different time horizons and sentiment regimes. Wavelet coherence is particularly valuable for analysing ESG and non-ESG futures in relation to uncertainty indices, as shocks such as geopolitical crises or tariff announcements may generate immediate but short-term effects, or alternatively, long-lasting and persistent consequences.

Another related literature examines connections and volatility spillovers across sustainability-related financial markets. (Zhang et al., 2022) studied spillovers among ESG indices, renewable energy stocks, green bonds, and carbon emission futures, finding that carbon markets often act as transmitters of volatility, while green bonds tend to be receivers. (Shaik and Rehman, 2023) analysed S&P ESG indices across regions using DCC-GARCH models, concluding that volatility connectedness varies by geography and intensifies during crises. In addition, wavelet-based and quantile-based methods have been used to document that ESG assets are deeply integrated with global financial cycles, though with horizon-specific differences. These findings suggest that ESG instruments are not isolated from traditional markets; rather, they participate actively in volatility networks. Futures, given their leverage and liquidity, are particularly likely to transmit shocks, but systematic evidence is still missing.

Three clear gaps emerge from this review. First, most studies analyse ESG indices or equities, with little attention to ESG futures despite their growing liquidity and institutional use. Second, while volatility modelling with GARCH is common, relatively few studies combine it with time-frequency methods such as wavelet coherence, which can uncover horizon-dependent dynamics. Third, existing analyses rarely integrate multiple types of uncertainty indices (VIX, EPU, GPR) in a comparative framework, even though each captures distinct sources of risk.

The present study addresses these gaps by applying wavelet coherence and GARCH models to ESG and non-ESG stock index futures over the period 2021–2025. It investigates their co-movement with VIX, GPR, and EPU across short-, medium-, and long-term horizons and assesses how major events such as the war in Ukraine and U.S. tariff shocks influence conditional variances. By doing so, it provides new evidence on whether ESG futures offer differentiated risk-return dynamics or whether they remain tightly coupled with conventional markets.

Based on the literature and preliminary evidence, the study proposes the following hypotheses:

H1. ESG futures are highly correlated with their conventional counterparts but show distinct reactions to uncertainty shocks.

H2. The VIX index has a stronger and more consistent relationship with futures returns (both ESG and non-ESG) compared to GPR and EPU.

H3. ESG futures exhibit lower volatility persistence compared to traditional futures, indicating greater resilience.

H4. Major geopolitical and economic events (e.g., Ukraine war, U.S. tariffs) significantly increase volatility in both ESG and non-ESG futures, but ESG futures respond differently in intensity or timing.

This research contributes to the debate on sustainable finance by providing empirical evidence on the behaviour of ESG derivatives under uncertainty. It offers insights for investors, policymakers, and risk managers interested in understanding whether ESG instruments can enhance market stability or whether they remain closely tied to broader market risks.

3. Data

The financial series included in the study are presented in Table 1. The data includes: three ESG stock indices futures: STOXX Europe 600 ESG-X Index Futures (FSEG), EURO STOXX 50 ESG Index Futures (FSSX), E-mini S&P 500 ESG Futures (ESG), three corresponding stock indices futures: STOXX Europe 600 Index Futures (FXXP), EURO STOXX Europe 50 Index Futures (STTX), E-mini S&P 500 Futures (ES) and three measures of uncertainty: CBOE volatility index (VIX), geopolitical risk index (GPR) and economic policy uncertainty index (EPU). CBOE volatility index (VIX), geopolitical risk index (GPR) and economic policy uncertainty index (EPU) were used as the measures of uncertainty. VIX measures the implied volatility of the S&P 500 index over the next 30 days. It reflects investor expectations about future market volatility. GPR is developed by Federal Reserve researchers (Caldara and Iacoviello, 2022). It measures the geopolitical risk by analysing the frequency of newspaper articles containing terms related to geopolitical tension. EPU proposed by (Baker et al., 2016) is based on the frequency of newspaper articles containing terms related to the economy, policy, and uncertainty.

The data were retrieved from LSEG Workspace based on the Partnership Agreement between University of Gdansk and LSEG. The research covered the period 02.01.2021−23.05.2025 (1145 daily 5-days observations). Analyses were performed using StataSE and R Statistical Software. To conduct wavelet analysis, the R package biwavelet was used.

4. Methodology

First, the unit root ADF-GLS test (Elliott et al., 1996), the Phillips-Perron unit root test (Phillips and Perron, 1988) and the stationarity KPSS test (Kwiatkowski et al., 1992) were performed to test for the order of integration of financial series used in the study. Next, the descriptive statistics, skewness, and kurtosis test (D'Agostino & Belanger, 1990), and Spearman correlation coefficients for logarithmic rates of return were calculated.

To further explore the dependence structure, additional analyses were conducted. To confirm the conclusions about the correlation between returns on contracts and indices, Spearman correlation coefficients were first calculated within decile-based return intervals for the VIX, EPU, and GPR indices (0–10th, 10–20th, …, 90–100th percentiles). This method allows an assessment of whether the strength of dependence varies across different parts of the return distribution, including extreme negative and positive market conditions. Secondly, rolling Spearman rank correlations were used to examine how the relationship between futures returns and uncertainty indicators evolves over time. Spearman coefficients were calculated over 30-, 60-, and 120-day windows. Rolling windows enable the correlation structure to adapt over time and reduce the influence of outliers. The rolling analysis provides insights into how the dependence fluctuates around major market events and periods of increased volatility.

Additionally, to analyse whether the correlation is stable during the research period, the wavelet coherence between log returns of futures prices and log returns of uncertainty indices was performed with the R wtc function. (Tarik et al., 2021).

The wavelet coherence between x_t and y_t was defined by (Grinsted et al., 2004) as an extension of the (Torrence and Compo, 1998) approach as follows:

Table 1
The financial series included in the study.

Index type	Ticker	Time series name	Source
ESG Indices futures	$FSEG_t$	STOXX® Europe 600 ESG-X Index Futures	LSEG
	$FSSX_t$	EURO STOXX 50® ESG Index Futures	LSEG
	ESG_t	E-mini S&P 500 ESG Futures (CME)	LSEG
Stock Indices futures	$FXXP_t$	STOXX® Europe 600 Index Futures	LSEG
	$STTX_t$	EURO STOXX Europe 50 Index Futures	LSEG
	ES_t	E-mini S&P 500 Futures (CME)	LSEG
Indices	VIX_t	CBOE VIX Index	LSEG
	GPR_t	Geopolitical Risk (GPR) Index	https://www.matteoiacoviello.com/gpr.htm
	EPU_t	Economic Policy Uncertainty Index for United States	FRED

Note. 1145 daily 5-day observations, 04.01.2021−23.05.2025, FRED – Federal Reserve Economic Data, LSEG – London Stock Exchange Group; the data were retrieved from LSEG Workspace based on the Partnership Agreement between the University of Gdansk and LSEG.
Source: the author's own research.

$$R^2(\tau,s) = \frac{\left|S\left(s^{-1}W_{xy}(\tau,s)\right)\right|^2}{S\left(s^{-1}\left|W_x(\tau,s)\right|^2\right)S\left(s^{-1}\left|W_y(\tau,s)\right|^2\right)} \tag{1}$$

where: $W_{xy}(\tau,s) = W_x(\tau,s)W_y^*(\tau,s)$ is the cross-wavelet transform of two series x_t and y_t, $W_x(\tau,s) = \int_{-\infty}^{\infty} x(t)\widetilde{\psi}^*_{\tau,s}(t)dt$, $W_y(\tau,s) = \int_{-\infty}^{\infty} y(t)\widetilde{\psi}^*_{\tau,s}(t)dt$ are the continuous wavelet transform of time series x_t and y_t, s is the scaling factor that defines the wavelet's length, τ ist the translation parameter determining the wavelet location in time, $\widetilde{\psi}^*_{\tau,s}(t) = \frac{1}{\sqrt{|s|}}\psi\left(\frac{t-\tau}{s}\right)$ is the complex conjugate function of $\psi^*_{\tau,s}(t)$, $\widetilde{\psi}$ is determined via scaling and shifting the mother wavelet ψ, s, $\tau \in R$, $s \neq 0$, S is the smoothing parameter, $0 < R^2(\tau,s) \leq 1$ (Aladwani, 2023; Goupillaud et al., 1984; Kilic et al., 2022).

The wavelet coherence $R^2(\tau,s)$ indicate what is the correlation between x_t and y_t localized in time and frequency. In the previous version of the R wtc function, to differentiate between positive and negative correlations, the method proposed by (Torrence and Compo, 1998) and (Grinsted et al., 2004) was applied. However, (Tarik et al. 2021) argue that because the traditional method can lead to a decrease in power at lower periods, the new version of R library employs (Liu et al., 2007) or (Veleda et al., 2012) bias correction methods. In the paper, the Morlet-type function ψ was used as the mother wavelet, and 1000 Monte Carlo randomisations were performed.

As a robustness test, the wavelet coherence was additionally calculated using the Paul wavelet function. The coherence matrices derived from the Paul wavelet were then compared to those from the Morlet wavelet. Both matrices were resized to the same dimensions, after which the Pearson correlation coefficient was computed for their corresponding elements. This method allowed for understanding how similar the coherence structures generated by the two wavelet types are.

Next, to model the volatility of daily logarithmic returns on futures prices, the univariate GARCH (Bollerslev, 1986) was used:

$$h_t = \alpha_0 + \sum\nolimits_{i=1}^{m} \alpha_i \varepsilon^2_{t-i} + \sum\nolimits_{j=1}^{n} \beta_j h_{t-j} \tag{2}$$

where: h_t – conditional variance of logarithmic returns on futures, ε_t – random error, $\varepsilon_t \sim iid(0,h_t)$, α_0,α_i, β_j – structural parameters such as α_0, $\alpha_i,\beta_j > 0$ and $\sum_{i=1}^{m}\alpha_i + \sum_{j=1}^{n}\beta_j < 1$.

Finally, the univariate GARCH-X model was employed to examine whether market volatility, as measured by VIX, affects the conditional volatility of futures returns:

$$h_t = \alpha_0 + \gamma_1 x_t + \sum\nolimits_{i=1}^{m} \alpha_i \varepsilon^2_{t-i} + \sum\nolimits_{j=1}^{n} \beta_j h_{t-j} \tag{3}$$

where: x_t – exogenous variable $\Delta lVIX_t$, α_0,α_i, β_j, γ_1 – structural parameters such as $\alpha_i,\beta_j > 0$ and $\sum_{i=1}^{m}\alpha_i + \sum_{j=1}^{n}\beta_j < 1$.

5. Results

Fig. 1 displays the levels of time series used in the study, and Fig. 2 shows the corresponding logarithmic rates of returns. As demonstrated on plots, the log daily series seem to be non-stationary. That assumptions are supported by the results of the unit root and stationarity tests for logarithmic rates of returns presented in Table 2. Estimated statistics for ADF-GLS test indicate that the null hypothesis of a unit root can be rejected for most variables. It cannot be rejected for $\Delta lVIX_t$ and $\Delta lGPR_t$. However, according to the Phillips-Perron test, the null hypothesis of a unit root can be rejected for all the variables. Results from KPSS test indicate that the null hypothesis of both level and trend stationary returns cannot be rejected.

Descriptive statistics and results from the skewness and kurtosis tests for the normality of logarithmic rates of return are presented in Table 3. For log futures returns, all means are positive during the period. The returns for all STOXX futures are negatively skewed and all variables are leptokurtic. These findings are supported by the results of (D'Agostino & Belanger, 1990) test. Based on kurtosis alone, the hypothesis of normality is also rejected for all variables. However, the skewness for $\Delta lESG_t$ and ΔlES_t are not significantly different from the skewness of a normal distribution. Despite that, the null hypothesis of normality can be rejected for all variables. In the case of uncertainty measures, according to descriptive statistics and test results, all variables are both negatively skewed and leptokurtic.

Fig. 3 presents the Spearman correlation coefficients between logarithmic rates of returns considered in the study. The ESG and corresponding non ESG futures are highly positively correlated. The coefficients are 0.996, 0.910 and 0.975 for the pairs FSEG-FXXP, FSSX-STTX and ESG-ES, respectively. In the case of correlation between futures contracts returns and uncertainty measures there is a negative correlation significant for VIX index. There is no significant correlation observed between futures returns and other indices – namely GPR and EPU. This suggests that the pricing of contracts is linked to the volatility of financial markets but is not strongly correlated with geopolitical events.

The results of additional Spearman correlation analysis are shown in Table 4. Reported values are the Spearman rank correlation coefficients between logarithmic rates of return for indices and futures, calculated within each percentile group of VIX, EPU and GPR, respectively. Fig. 4 shows the correlation coefficients with the log returns of VIX. The link between futures and VIX returns heavily depends on the volatility regime: the more stress there is, the more reliably futures fall when the VIX rises. The correlation is stronger in the upper percentiles, indicating a greater dependency during periods of high uncertainty.

Fig. 5 depicts rolling Spearman coefficients calculated over 30-, 60-, and 120-day windows. The results indicate that correlations

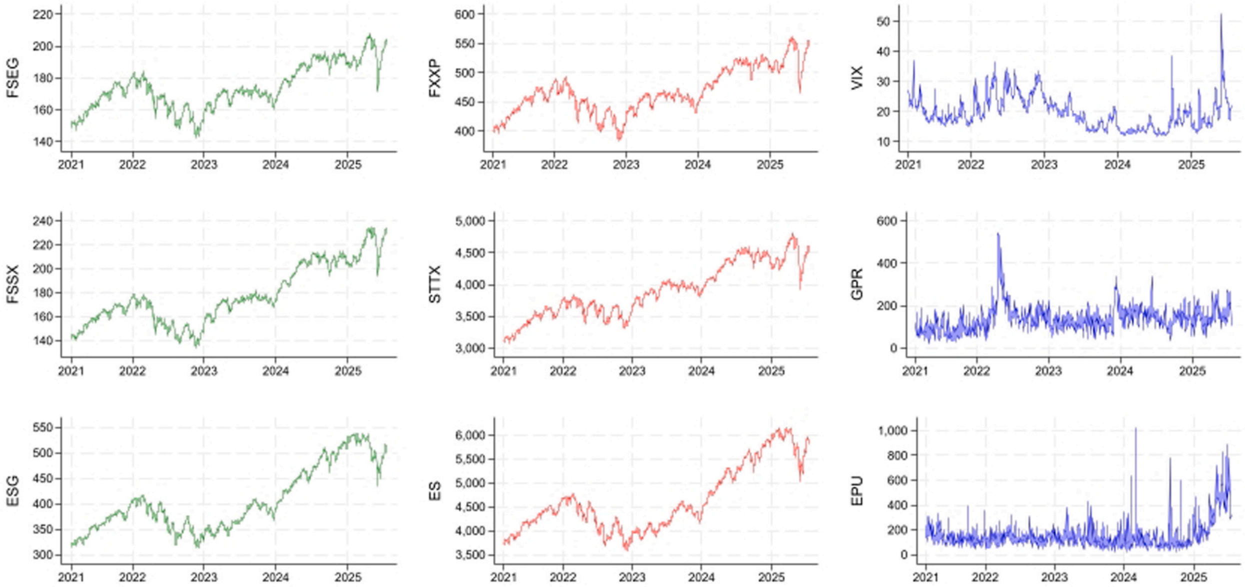

Fig. 1. Financial series levels. Source: the author's own research.

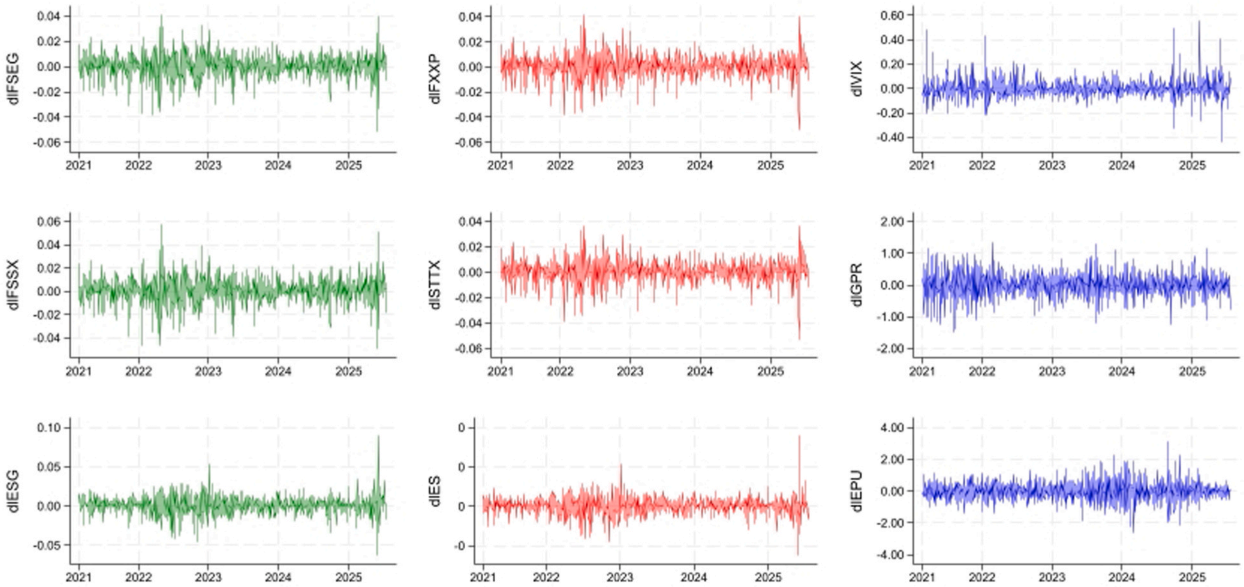

Fig. 2. Logarithmic rates of returns. Source: the author's own research.

Table 2
Unit root and stationarity tests for logarithmic rates of returns.

	ADF-GLS				PP				KPSS			
	level	lags	trend	lags	level	lags	trend	lags	level	lags	trend	lags
$\Delta lFSEG_t$	−3.487***	18	−5.175***	18	−34.667***	6	−34.652***	6	0.042	7	0.043	7
$\Delta lFSSX_t$	−1.984**	21	−3.857***	18	−35.839***	6	−35.824***	6	0.056	7	0.046	7
$\Delta lESG_t$	−1.955**	22	−3.514***	22	−34.218***	6	−34.203***	6	0.066	7	0.070	7
$\Delta lFXXP_t$	−3.850***	18	−5.503***	18	−34.728***	6	−34.712***	6	0.042	7	0.042	7
$\Delta lSTTX_t$	−2.516**	21	−4.550***	18	−35.032***	6	−35.024***	6	0.040	7	0.026	7
ΔlES_t	−1.582	22	−3.244**	22	−34.082***	6	−34.067***	6	0.073	7	0.065	7
$\Delta lVIX_t$	−1.373	22	−2.706*	22	−35.553***	6	−35.539***	6	0.027	7	0.015	7
$\Delta lGPR_t$	−0.798	22	−1.086	22	−72.328***	6	−72.288***	6	0.008	7	0.006	7
$\Delta lEPU_t$	−1.658	22	−3.222***	22	−84.584***	6	−84.568***	6	0.016	7	0.005	7

Note. The symbols *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.
Source: the author's own research.

Table 3
Descriptive statistics and tests form normality of logarithmic rates of return.

	Mean	Standard dev.	Minimum	Maximum	Skewness	Kurtosis	pr(S)	pr(K)	N
$\Delta lFSEG_t$	0.0003	0.0093	−0.0509	0.0406	−0.5169	6.2093	0.00	0.00	108.71***
$\Delta lFSSX_t$	0.0004	0.0109	−0.0491	0.0573	−0.3003	5.8318	0.00	0.00	78.45***
$\Delta lESG_t$	0.0004	0.0111	−0.0620	0.0896	0.0405	8.8475	0.57	0.00	118.40***
$\Delta lFXXP_t$	0.0003	0.0092	−0.0506	0.0411	−0.5621	6.4163	0.00	0.00	118.47***
$\Delta lSTTX_t$	0.0003	0.0088	−0.0536	0.0370	−0.6400	6.7491	0.00	0.00	135.13***
ΔlES_t	0.0004	0.0109	−0.0612	0.0896	0.0241	9.1095	0.74	0.00	121.87***
$\Delta lVIX_t$	−0.0002	0.0767	−0.4424	0.5541	1.1314	11.2689	0.00	0.00	272.72***
$\Delta lGPR_t$	0.0000	0.3794	−1.4993	1.3573	−0.0828	3.8617	0.00	0.00	18.37***
$\Delta lEPU_t$	0.0003	0.5866	−2.6351	3.1390	0.1645	4.9275	0.02	0.00	48.75***

Note. $\Delta l(*)_t$ – logarithmic rate of return, pr(S) – p-value for skewness test for normality, pr(K) – p-value for kurtosis test for normality, N – statistic for D'Agostino & Belanger (1990) test for normality, 1114 daily observations.
Source: the author's own research.

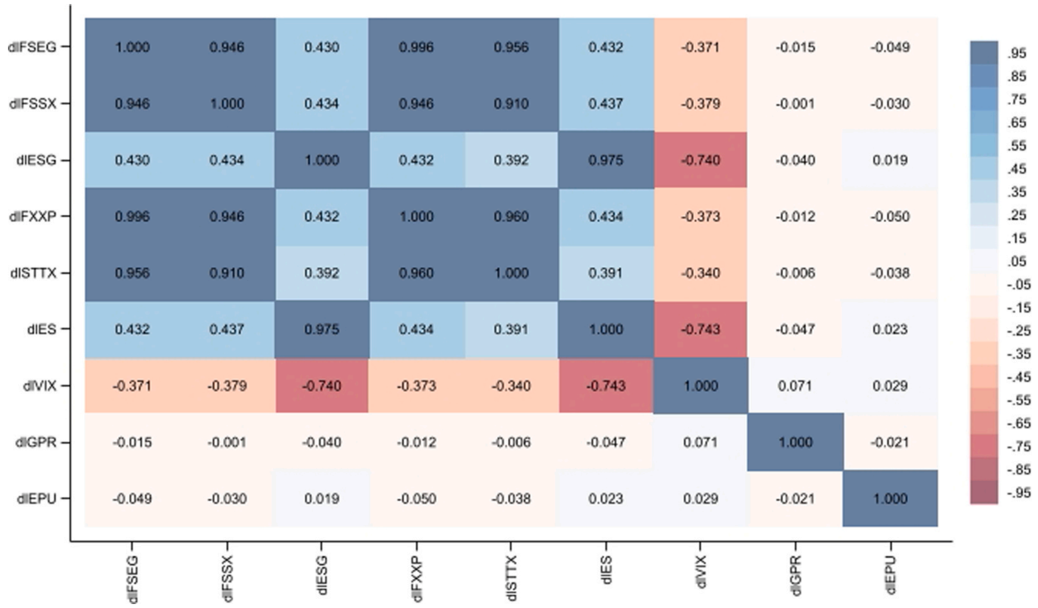

Fig. 3. Spearman Correlation for logarithmic rates of returns. Source: the author's own research.

between futures returns and uncertainty indicators (VIX, GPR, EPU) depend on the estimation horizon. Shorter windows produce volatile, frequently changing correlations, whereas longer windows exhibit smoother, more consistent patterns, aligning with medium- to long-term dependence structures. The most stable relationships are seen with VIX, reflecting its role as a forward-looking measure of

Table 4
Spearman Correlation for logarithmic rates of returns.

	P0−10	P10−20	P20−30	P30−40	P40−50	P50−60	P60−70	P70−80	P80−90	P90−100
VIX_t										
$\Delta lFSEG_t$	−0.060	−0.303***	−0.356***	−0.421***	−0.296***	−0.431***	−0.401***	−0.561***	−0.296***	−0.371***
$\Delta lFSSX_t$	−0.113	−0.322***	−0.262***	−0.395***	−0.375***	−0.458***	−0.407***	−0.556***	−0.302***	−0.382***
$\Delta lESG_t$	−0.326***	−0.597***	−0.664***	−0.780***	−0.764***	−0.799***	−0.786***	−0.864***	−0.721***	−0.849***
$\Delta lFXXP_t$	−0.044	−0.306***	−0.353***	−0.419***	−0.295***	−0.450***	−0.394***	−0.552***	−0.302***	−0.381***
$\Delta lSTTX_t$	−0.083	−0.243***	−0.329***	−0.328***	−0.246***	−0.453***	−0.343***	−0.506***	−0.242***	−0.371***
ΔlES_t	−0.375***	−0.580***	−0.657***	−0.759***	−0.806***	−0.817***	−0.801***	−0.875***	−0.716***	−0.842***
EPU_t										
$\Delta lFSEG_t$	−0.002	0.071	0.005	−0.063	−0.068	−0.170*	−0.185**	0.117	−0.142	−0.071
$\Delta lFSSX_t$	−0.003	0.096	0.048	−0.068	−0.075	−0.201**	−0.227**	0.147	−0.120	−0.032
$\Delta lESG_t$	0.047	−0.023	0.084	0.106	0.065	−0.029	−0.195**	0.168*	−0.039	0.013
$\Delta lFXXP_t$	0.011	0.066	0.003	−0.071	−0.071	−0.182*	−0.199**	0.110	−0.153	−0.070
$\Delta lSTTX_t$	0.040	0.065	−0.005	−0.056	−0.125	−0.158*	−0.183**	0.097	−0.140	−0.041
ΔlES_t	0.031	−0.038	0.096	0.082	0.070	−0.025	−0.175*	0.139	−0.025	0.003
GPR_t										
$\Delta lFSEG_t$	0.140	−0.051	−0.135	−0.126	0.096	0.051	0.033	0.016	0.147	−0.060
$\Delta lFSSX_t$	0.092	−0.068	−0.122	−0.078	0.110	0.084	0.056	0.061	0.111	−0.030
$\Delta lESG_t$	0.101	−0.016	−0.032	−0.012	0.024	−0.133	−0.031	0.108	−0.053	−0.118
$\Delta lFXXP_t$	0.151	−0.044	−0.141	−0.115	0.095	0.038	0.044	0.021	0.144	−0.064
$\Delta lSTTX_t$	0.156*	−0.019	−0.176*	−0.135	0.137	0.042	0.084	0.073	0.143	−0.110
ΔlES_t	0.107	−0.040	−0.033	0.008	0.009	−0.151	−0.057	0.102	−0.060	−0.135

Note. $\Delta l(*)_t$ – logarithmic rate of return, P0–10, P10–20, …, P90–100 denote percentile intervals of the distribution of VIX_t, EPU_t and GPR_t respectively. Reported values are Spearman rank correlation coefficients between logarithmic rates of return for indices and futures, calculated within each percentile group, the symbols *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.
Source: the author's own research.

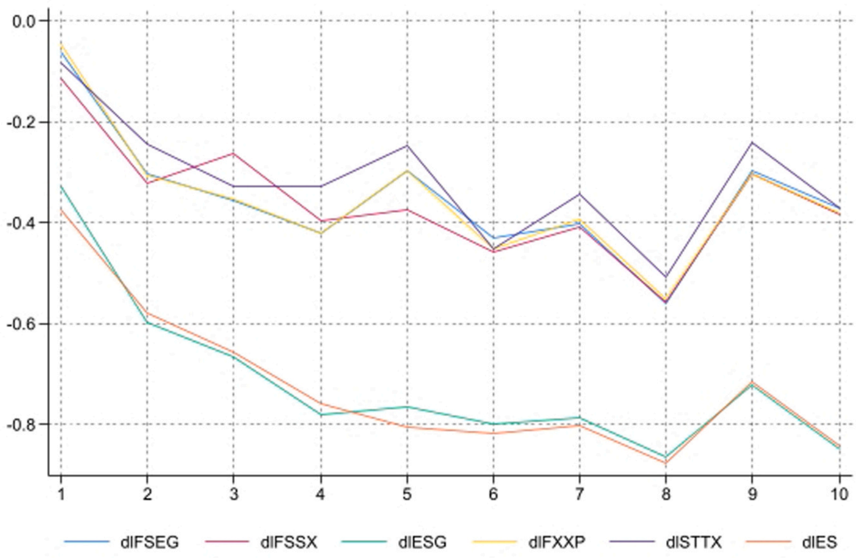

Fig. 4. Spearman Correlation for logarithmic rates of returns of VIX and futures. Source: the author's own research.

Fig. 5. Rolling Spearman Correlation for logarithmic rates of returns. Source: the author's own research.

overall market risk aversion (Whaley, 2000). Correlations with GPR are weaker and episodic, consistent with the jump-like nature of geopolitical shocks (Caldara & Iacoviello, 2022), while EPU exhibits moderate persistence, consistent with its slower impact on financial markets (Baker et al., 2016).

Next, to examine the linkages localized in time and frequency, the wavelet coherence analysis was performed. Fig. 6 presents the

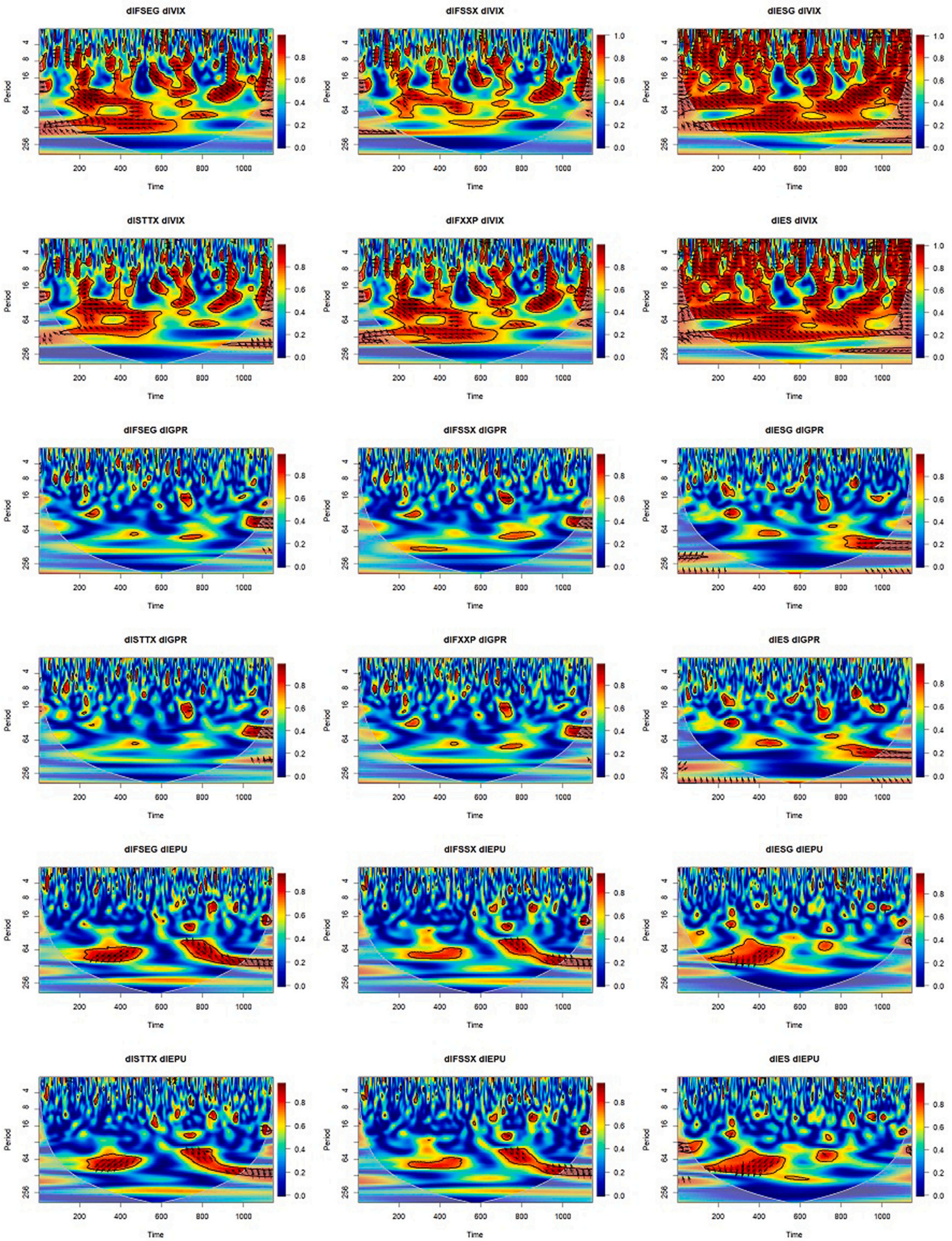

Fig. 6. Wavelet coherency between log returns of stock futures prices, VIX, GPR and EPU indices. Source: the author's own research.

wavelet coherence between log returns of stock futures prices, the VIX, GPR, and EPU indices. The X-axis represents time t (1114 daily observations), while the Y-axis represents period – the scale (ranging from 0 to 256 days), where the ranges 0–16, 16–64, and 64–256 can be interpreted as short-, medium-, and long-term time scales, respectively. The heatmap on the plot shows the strength of the co-movement between the variables (0 – dark blue – minimal series dependence, 1 – dark red – high series dependence). The arrows indicate the phase relationship: an arrow pointing to the right indicates positive co-movements (in-phase), while an arrow pointing to the left indicates negative co-movements (out-of-phase connections). The right-up or left-down arrows indicate that the first signal leads the second, and the right-down or left-up arrows that the second signal leads the first, respectively. Wavelet coherence analysis does not indicate causality – it only reflects the strength and nature of the relationship (Kilic et al., 2022).

As indicated by the Spearman correlation analysis, the coherence between log futures returns and the VIX index is significant at various frequencies, time scales and various sample episodes. The coherence is the strongest for the pairs ESG-VIX and ES-VIX which may be caused by the construction of the VIX index based on S&P 500 option-implied volatility. The arrows point left or left-up, indicating that VIX leads ESG and ES, and that the connection is out of phase. The relation between futures based on the STOXX indices, both ESG and non-ESG, and VIX is mainly visible in 2022 (observations 200–500) as well as in the later period after 2024, both in short- and medium-term time scales.

The strong high-frequency coherence among ESG, ES futures, and VIX indicates that these markets primarily respond to short-term volatility shocks driven by news and sudden shifts in risk sentiment. These co-movements occur during periods of uncertainty, such as heightened equity volatility in 2022, unexpected macroeconomic news, or geopolitical stress, leading to quick risk-off responses. In contrast, the coherence between STOXX futures and the VIX at medium-term frequencies indicates slower transmission of US volatility shocks to European markets and more persistent cross-market adjustments.

Regarding the relationship between futures and GPR, there is little significant coherence over the period. This suggests that geopolitical risk does not trigger price reactions in futures markets. Finally, the wavelet coherence between futures and EPU log returns is noticeable at medium-term time scales (64–128 days), but only from 2022 onwards, and after 2024. It indicates that economic policy uncertainty results in slower fundamental adjustments than the VIX and influences futures markets through gradual macroeconomic channels.

The results of a robustness test on wavelet coherence are presented in Table 5. The Pearson correlation coefficients show that the robustness of wavelet coherence estimates varies significantly across risk factors. Coherence patterns involving VIX exhibit relatively high stability in the choice of mother wavelet, with correlations typically ranging from 0.47 to 0.59. Conversely, coherence measures related to geopolitical risk (GPR) exhibit noticeably lower robustness, with correlations ranging from 0.34 to 0.50, indicating that the detection of GPR-driven co-movements depends more on the chosen wavelet. The EPU index falls in between, with correlations from 0.40 to 0.45, suggesting moderate sensitivity to the wavelet configuration. Overall, these findings indicate that the reliability of coherence estimates is highest for VIX, moderate for EPU, and lowest for GPR.

The main estimation and validation results of GARCH models for logarithmic rates of futures returns are gathered in Table 6. They are correctly specified, as the Ljung-Box test null hypothesis is not rejected up to the 20th order as far as the hypothesis of an integrated GARCH is rejected (see the estimates of *IG* test statistics). The behaviour of the conditional variances is illustrated in Fig. 7. Periods of increased conditional variances are observed. The volatility of futures prices has been reacting to the situation on financial markets as well as geopolitical events like the war in Ukraine. Fig. 8 shows the behaviour of conditional variances after the outbreak of the war. Increased volatility was observed from the end of 2021 to mid−2023. It has exhibited a correlation with surging inflation and successive interest rate hikes in the US and Europe. However, as shown in Fig. 9, the largest increase in conditional variances occurred in April 2025. On "Liberation Day," April 2, 2025, Trump signed a decree establishing a 10% base tariff on imports from all countries (effective April 5), along with higher "reciprocal tariffs" (effective April 9), reaching, for example, 20% for the EU and 34% for China. The conditional variances for $\Delta lESG_t$ and ΔlES_t reached their maximum values on April 11 and 10, respectively. European markets also responded with increased volatility during this period, reaching their highest levels of volatility in the period analysed.

The main estimation and validation results of GARCH-X models are presented in Table 7. All the models are correctly specified. Including the additional variable in the variance equation improved the AIC. A positive and statistically significant coefficient of the $\Delta lVIX_t$ in the variance equation confirms that global financial uncertainty is transmitted to the volatility of ESG futures returns. Changes in this index influence fluctuations in market volatility.

6. Conclusions

This study set out to investigate the relationship between ESG and conventional futures and their interactions with measures of uncertainty (VIX, GPR, EPU) in the period 2021–2025. By applying wavelet coherence and GARCH modelling, it explored both the

Table 5
Correlation Between Paul and Morlet Wavelet Coherence.

	$\Delta lFSEG_t$	$\Delta lFSSX_t$	$\Delta lESG_t$	$\Delta lFXXP_t$	$\Delta lSTTX_t$	ΔlES_t
$\Delta lVIX_t$	0.583	0.570	0.473	0.589	0.594	0.496
$\Delta lGPR_t$	0.376	0.421	0.501	0.378	0.339	0.486
$\Delta lEPU_t$	0.425	0.447	0.425	0.421	0.400	0.454

Note. Reported values are Pearson correlation coefficients between Paul and Morlet squared wavelet coherence.
Source: the author's own research.

Table 6
Estimation and validation results of the GARCH models.

	$\Delta lFSEG_t$	$\Delta lFSSX_t$	$\Delta lESG_t$	$\Delta lFXXP_t$	$\Delta lSTTX_t$	ΔlES_t
Variance equation						
$\widehat{\alpha}_0$	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***
$\widehat{\alpha}_1$	0.171***	0.166***	0.067***	0.178***	0.163***	0.099***
$\widehat{\alpha}_2$			0.043*			
$\widehat{\beta}_1$	0.716***	0.712***	0.859***	0.698***	0.707***	0.875***
Validation						
IG	21.650***	24.220***	13.100***	22.870***	29.130***	9.920***
LB(5)	2.555	4.750	4.488	2.488	3.441	3.710
LB(20)	14.006	17.721	17.953	13.441	13.180	21.599
$LB_2(5)$	3.964	1.557	11.582	3.991	3.655	10.124
$LB_2(20)$	7.843	5.755	28.729	7.997	8.682	27.767
AIC	−7638.749	−7249.685	−7281.695	−7650.364	−7742.805	−7332.542

Note. The symbols *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively, IG – Wald test statistic, IGARCH vs. GARCH, Ljung Box test was calculated for standardized innovations (LB(5), LB(20)) and squared standardized innovations ($LB_2(5)$, $LB_2(20)$).
Source: the author's own research.

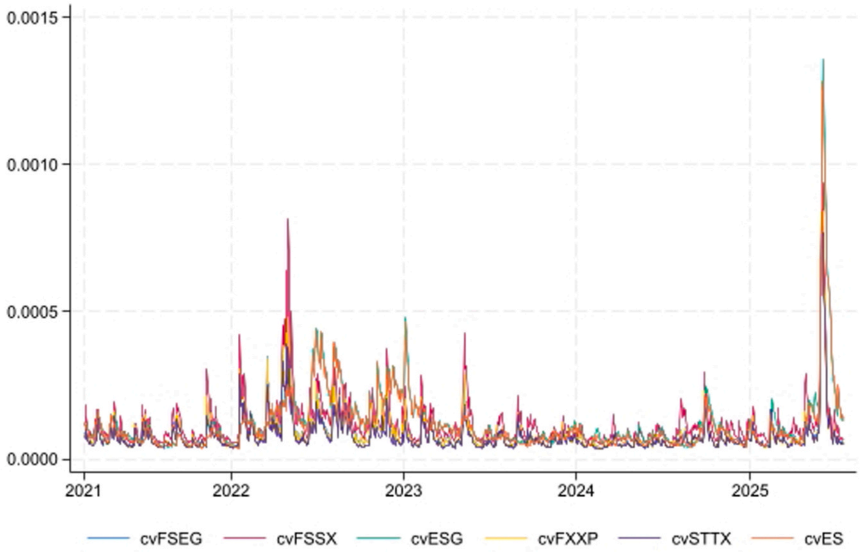

Fig. 7. Conditional variance from GARCH models. Source: the author's own research.

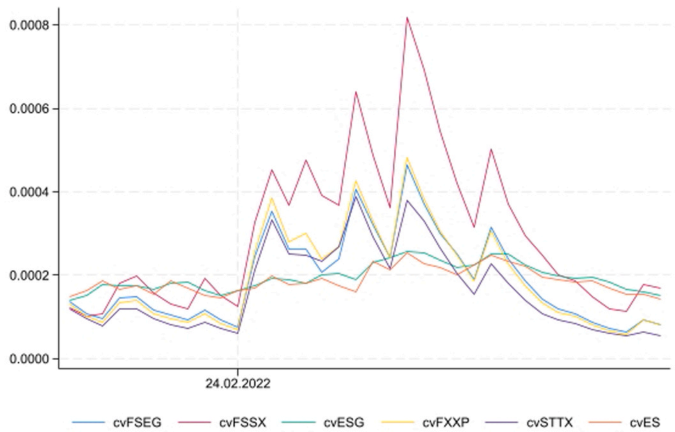

Fig. 8. Conditional variance from GARCH models – the Russian invasion of Ukraine. Source: the author's own research.

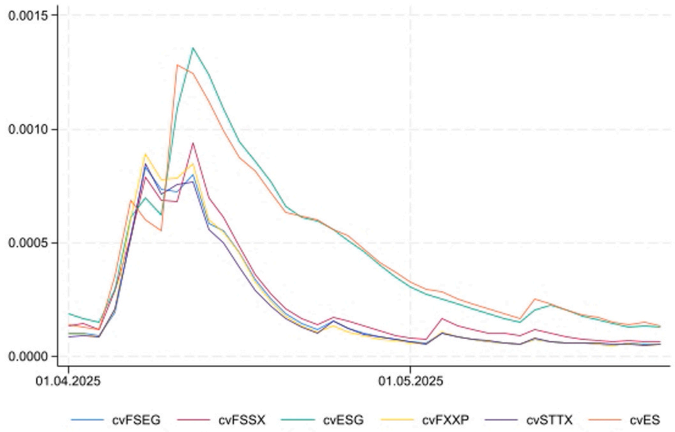

Fig. 9. Conditional variance from GARCH models – the 2025 U.S. tariffs. Source: the author's own research.

Table 7
Estimation and validation results of the GARCH-X models.

	$\Delta lFSEG_t$	$\Delta lFSSX_t$	$\Delta lESG_t$	$\Delta lFXXP_t$	$\Delta lSTTX_t$	ΔlES_t
Variance equation						
$\widehat{\gamma}_1$	8.843***	8.579***	10.393***	8.911***	9.038***	11.112***
$\widehat{\alpha}_0$	−11.762***	−11.468***	−11.843***	−11.742***	−11.869***	−11.476***
$\widehat{\alpha}_1$	0.141***	0.142***		0.144***	0.122***	0.044*
$\widehat{\alpha}_2$			0.062**			0.035
$\widehat{\alpha}_3$			0.135***			0.070**
						0.153***
$\widehat{\beta}_1$	0.720***	0.729***	0.696***	0.712***	0.736***	0.525***
Validation						
IG	52.870***	49.230***	40.740***	53.210***	66.180***	45.070***
LB(5)	3.505	6.069	5.721	3.421	5.444	4.595
LB(20)	14.892	17.782	20.306	13.808	15.151	21.524
$LB_2(5)$	7.246	1.450	4.970	6.545	7.098	6.235
$LB_2(20)$	21.889	13.526	27.552	21.762	21.750	22.818
AIC	−7714.347	−7294.849	−7359.597	−7709.732	−7810.566	−7412.872

Note. The symbols *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively, IG – Wald test statistic, IGARCH vs. GARCH, Ljung Box test was calculated for standardized innovations (LB(5), LB(20)) and squared standardized innovations ($LB_2(5)$, $LB_2(20)$).
Source: the author's own research.

time-frequency dynamics of co-movement and the persistence of volatility under normal conditions and during episodes of heightened market stress.

The results provide several key insights. First, ESG futures display a strong positive correlation with their conventional counterparts, confirming H1, but the strength and direction of their responses to uncertainty shocks differ depending on the index and time horizon. Second, the findings support H2, showing that the VIX index exhibits the most robust and consistent relationship with both ESG and non-ESG futures, while the influence of GPR and EPU is weaker and less stable. Third, evidence from the GARCH models partly supports H3, as ESG futures exhibit differences in volatility persistence compared to traditional contracts, suggesting some degree of resilience in turbulent periods. Finally, major geopolitical and economic shocks, such as the war in Ukraine and the 2025 U.S. tariffs, had a pronounced impact on volatility, validating H4 and highlighting the sensitivity of both ESG and non-ESG futures to global events.

The empirical results reveal that the relationship between futures returns and uncertainty is not uniform but depends on the nature of the underlying shock. In particular, the role of ESG futures varies across financial, geopolitical, and policy-driven sources of uncertainty, which provides a more nuanced interpretation of their behaviour under stress conditions. In the case of financial market uncertainty, as proxied by the VIX, both ESG and conventional futures exhibit strong and persistent co-movement across time and frequency domains. This finding suggests that ESG futures are fully exposed to global risk sentiment and respond in a synchronised manner with traditional assets. Therefore, the observed dynamics cannot be interpreted as evidence of safe haven properties. Instead, they reflect a common risk-transmission mechanism in which the VIX serves as a dominant channel for market-wide uncertainty. Under such conditions, ESG futures do not provide superior hedging benefits relative to conventional futures, as both asset classes are driven by the same underlying factor: investor risk aversion.

A different pattern emerges in the case of geopolitical risk (GPR). The results indicate limited and episodic coherence between futures returns and the GPR index, suggesting that geopolitical shocks do not systematically translate into price dynamics in either ESG or non-ESG futures markets. This may reflect the indirect and heterogeneous nature of geopolitical events, which affect markets through multiple transmission channels and with varying intensity across regions and sectors. As a consequence, ESG futures do not demonstrate a consistent hedging advantage in response to geopolitical risk, and their behaviour remains broadly aligned with conventional instruments.

For economic policy uncertainty (EPU), the relationship is more gradual and becomes visible primarily at medium-term frequencies. This indicates that policy-related shocks are absorbed through slower macroeconomic adjustment processes rather than immediate market reactions. While ESG futures may exhibit differences in the persistence and timing of volatility adjustments compared to conventional futures, these differences do not imply safe haven characteristics. Instead, they point to variations in how volatility propagates and dissipates over time.

Taken together, these findings suggest that ESG futures should not be classified as safe haven assets under different stress scenarios. Rather than providing protection against specific types of uncertainty, they remain embedded within the broader system of financial risk transmission. However, the observed differences in volatility persistence indicate that ESG futures may still offer a form of relative resilience, not by avoiding shocks, but by adjusting to them in a distinct manner. This distinction is crucial, as it shifts the interpretation of ESG instruments from protective assets to components of a differentiated, yet interconnected, market structure. This suggests that ESG futures should be understood not as alternative or protective assets, but as components of a differentiated yet integrated risk system.

The study makes several contributions to the literature on financial markets, uncertainty, and sustainable finance. First, it extends the growing body of work on ESG investments by focusing specifically on derivatives markets, an area that remains underexplored compared to equities and bonds. While most prior research has analysed ESG indices or funds, this paper investigates ESG futures, which are increasingly important for hedging and speculative strategies in institutional portfolios. Second, by employing wavelet coherence, the study provides novel evidence on the time-frequency co-movement between ESG and conventional futures and key measures of uncertainty (VIX, GPR, EPU). This approach reveals how relationships vary across short-, medium-, and long-term horizons, offering a more granular understanding than traditional correlation or regression analyses. Third, through GARCH modelling, the paper identifies differences in volatility persistence between ESG and non-ESG futures, shedding light on whether sustainable derivatives offer distinctive risk-return characteristics or resilience during crises. Fourth, the research is among the first to incorporate the effects of major geopolitical and policy shocks (including the war in Ukraine and U.S. tariff announcements in 2025) into the analysis of ESG derivatives markets. This enhances the relevance of the findings for both academics and practitioners by demonstrating how global uncertainty episodes shape volatility dynamics.

Taken together, these contributions position the study at the intersection of sustainable finance, econometrics, and uncertainty research. It advances our understanding of whether ESG futures behave as substitutes or complements to conventional instruments under stress, thereby offering implications for portfolio risk management rather than traditional diversification benefits.

Overall, the study contributes to the literature on sustainable finance by clarifying the extent to which ESG derivatives share or diverge from the risk-return dynamics of conventional futures. The results indicate that ESG instruments do not operate in isolation from broader market forces but may still offer differentiated volatility profiles that are relevant for risk management.

From a practical perspective, the findings have implications for investors and policymakers. Investors seeking diversification or hedging strategies should be aware that ESG futures, while highly correlated with traditional contracts, can behave differently under uncertainty, offering opportunities to adjust portfolio risk exposures. Policymakers and regulators, in turn, should recognise that the resilience of ESG instruments is not absolute, as global shocks continue to drive volatility across both sustainable and conventional markets.

The study has certain limitations. First, wavelet coherence analysis may be influenced by data frequency (daily versus weekly), as the scale-to-period conversion in the continuous wavelet transform depends directly on the sampling interval (Torrence and Compo, 1998). Therefore, future research should incorporate alternative data frequencies to assess the robustness of the identified time-frequency patterns. Second, while wavelet coherence captures co-movement, it does not establish causality. Future studies could employ more advanced models of dynamic dependence, such as DCC-MGARCH or other multivariate frameworks, to better identify transmission mechanisms between uncertainty and futures markets. Third, the 2025 tariff shocks analysed in this study likely represent short-lived, event-driven disruptions rather than long-term structural changes. As such, their implications for long-term market behaviour remain uncertain and require further investigation as more post-event data become available.

Importantly, future research should move beyond aggregate analysis and explicitly examine the underlying mechanisms driving the observed dynamics. In particular, it would be valuable to distinguish between investor-driven effects (e.g., ESG-related capital flows) and structural market characteristics (e.g., leverage, liquidity, or participant composition) as potential sources of the relative resilience observed in ESG futures. Moreover, further studies should explore whether ESG futures behave differently across distinct sources of uncertainty, such as financial market volatility, geopolitical risk, and economic policy uncertainty. A more granular, cross-sectional approach could help determine whether ESG instruments provide differentiated responses depending on the nature of the shock.

Finally, extending the analysis to additional ESG derivatives, incorporating cross-market spillovers, and integrating macroeconomic fundamentals would provide a more comprehensive understanding of how sustainable finance instruments interact with systemic uncertainty and global risk. Such extensions would help clarify whether ESG futures offer structurally distinct dynamics or remain primarily embedded within broader market risk transmission mechanisms.

CRediT authorship contribution statement

Magdalena Mosionek-Schweda: Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Marta Chylińska:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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